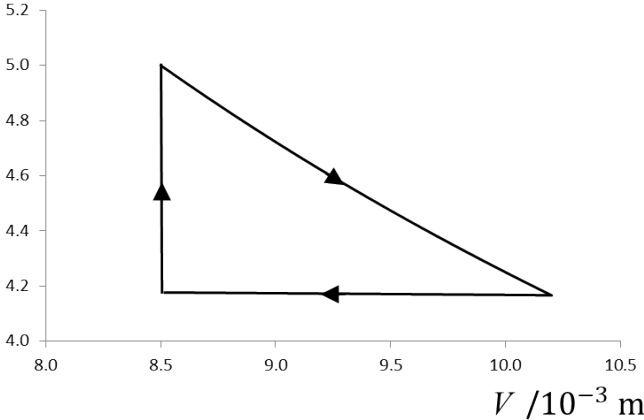


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	Substitution $n = \frac{pV}{RT} = \frac{(5 \times 10^5)(8.5 \times 10^{-3})}{(8.31)(285)}$ (1) $= 1.79 \text{ [mol]} \text{ (1)}$	1	1		2	2	
		(ii)	$N = N_A n = (6.02 \times 10^{23}) \times 1.79 \text{ ecf} = 1.08 \times 10^{24}$		1		1	1	
		(iii)	Substitution of p and V or k and T (1) Correct use of Nm or m in either: $p = \frac{1}{3} \rho \overline{c^2}$ or $pV = \frac{1}{3} Nm \overline{c^2}$ ecf (1) $c_{\text{rms}} = 471 \text{ [m s}^{-1}\text{]} \text{ (1)}$	1	1 1		3	2	
		(iv)	Force $= pA = (5.0 \times 10^5) \times 0.04 = 20\,000 \text{ [N]}$		1		1	1	
	(b)	(i)	Substitution e.g. $p = \frac{5.0 \times 10^5 \times 8.5 \times 10^{-3}}{10.2 \times 10^{-3}} \text{ (1) ecf on } n \text{ if}$ $pV = nRT$ used $p = 420 \text{ k[Pa]} \text{ (1)}$	1	1		2	2	
		(ii)	$\Delta U = 0 \text{ (1)}$ So by using the first law of thermodynamics $\Delta U = Q - W$ hence $Q = W = 773 \text{ [J]} \text{ (1)}$	1	1		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Work done = $[-]710$ [J] or area of triangle attempted (1) Total work done by the gas around cycle = $773 - 710 + 0 = 63$ [J] and $Q = W = 63$ [J] (1) Axes labelled with units (1) Correct closed triangle as shown (1) Treat arrows as neutral $p / 10^5 \text{ Pa}$  $V / 10^{-3} \text{ m}^3$			4	4	3	
			Question 5 total	4	7	4	15	13	0